Department of Economics

MSc Extended Essay Cover Sheet

CANDIDATE NUMBER *(This is a **5 digit number** found on LFY. It is **not** your 9 digit student number)*	**2**	**9**	**1**	**9**	**7**

COURSE CODE	**E**	**C**	**4**	**2**	**3**

TITLE	**The Effects of Schooling Mode on Parental Labor Market Outcomes**

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Double spaced, typed. Maximum 6,000 words (as a guideline this is 14-21 double spaced pages). Abstract, footnotes, references and appendices do not count toward the word count, provided such additions are brief and do not contain information that rightly belongs in the body of the essay. Equations are included in the word count and counted as the page equivalent (i.e., as the number of words that would occupy the same amount of space in text). Examiners are recommended to penalise excessively long essays (+ 10% of word count) by adjusting the marks by 6000/x, where x is the word count. Pages are to be numbered.*

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The Effects of Schooling Mode on Parental Labor Market Outcomes

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Abstract

This paper studies the effect of schooling mode on individual- and household-level labor market outcomes including total weekly hours worked, total weekly hours actually worked, and hours taken off per week. I merge data from the Basic Monthly Current Population Survey (CPS) with data on schooling mode from the COVID-19 School Data Hub to conduct my analysis. While I do not find any significance of state averaged share of time spent in in-person instruction on individual-level labor market outcomes of mothers and fathers, my results suggest that in-person and hybrid schooling modes significantly improve household-level labor market outcomes. I find that a shift from a 0% to 100% in-person schooling mode is commensurate to a 27.3 hour increase in total household hours worked and a 22.6 hour increase when this shift is made from hybrid instruction. When controlling for respective spouses' industry fixed effects, the positive effects remain significant but decrease to 8.8 additional hours worked per week for in-person schooling and 7.1 additional hours for hybrid schooling. While these findings are unable to unilaterally identify the incidence of the burden of virtual schooling mode on mothers as opposed to fathers, they provide substantial evidence of the negative impact of the lack of in-person schooling on parental household labor market outcomes.

Keywords: COVID-19, pandemic, United States, employment, labor market outcomes, school closures, schooling mode, gender equality

Word Count: 5,996

1 Introduction

The recession caused by the COVID-19 pandemic was fundamentally different from prior recessions. Economies closed to a standstill. [Goldin, 2022] As a result, the economic effects of the pandemic have been far-reaching. High levels of uncertainty about the direction of the economy were augmented by doubt about length of lockdowns, working from home, and virtual schooling—homes became workplaces, schools, and daycare centers. School closures and disruptions in children's learning not only caused significant learning losses for students, but also increased parental childcare burden.[Goldin, 2022]

Mothers, who remain as primary caretakers of children in the household, shouldered most of the burden caused by the closure of schools and daycare centers. This shift made it increasingly difficult for women to keep their jobs. [Collins et al., 2021a] Moreover, the pandemic has had a significantly greater negative impact upon the labor market outcomes of women than on those of men. Women were more likely to be laid off or furloughed and suffered from larger reductions in working hours. [Stantcheva, 2022] In light of the gender disparities exposed and exacerbated by the pandemic, it is important to consider how in-person schooling could help us remedy the gender gaps in the labor market.

In this paper, I explore how school re-openings throughout the 2020-21 school year affected parental labor market outcomes. I conduct my analysis at two distinct levels. I examine the effect of the share of time spent in different schooling modes within a state on individual- and household-level labor market outcomes.

To conduct this analysis, I combine data from the Basic Monthly Current Population Survey (CPS) with school learning model data from the COVID-19 School Data Hub (CSDH) for a set of 38 states. I begin by describing the changes in labor market outcomes across the 2020-21 school year by documenting statistics for the beginning (October 2020) and the end (May 2021) months of my analysis. I show that over the course of the 2020-21 school year, the shares of a given month that states spent in in-person schooling significantly increased.

A fundamental concern for my analysis is that schooling mode was not exogenously assigned to states. There could exist other state-level variables correlated with schooling mode and labor market outcomes, the omission of which would result in bias. I identify that

correctly specifying my estimating equation requires the inclusion of a monthly COVID-19 case rate control.

I conduct my analysis for three labor market variables: total hours worked, total hours actually worked, and hours taken off. I find no significant results of schooling mode on men's labor market outcomes. My results suggest that in-person schooling mode has a significant negative effect commensurate to a decrease in 1.2 hour of hours worked for women overall. However when restricting the sample to women with children, the treated population, this effect diminishes and becomes insignificant. I also observe a significant negative effect of -1.7 hours per week of in-person schooling mode on hours actually worked by mothers.

The negative effect of in-person schooling mode on hours worked at the individual level counters the causal mechanism hypothesized and documented in previous research. For instance, Garcia and Cowan (2022) show that school closures have had a significant effect on whether parents work full time and on the number of hours that parents work per week. They estimate that switching a county from entirely in-person to closed would result in the probability of full-time work decreasing by 3.8% and the hours worked of parents with school-aged children being reduced by 6.5% for mothers and 4.3% for fathers. [Garcia and Cowan, 2022] Similarly, Heggeness (2020) shows that the immediate response of working parents of school-age children to school closures resulted in mothers taking a significant amount of time off with no similar effect for fathers except for a small reduction in hours by approximately 0.53 per week. [Heggeness, 2020] Furthermore, in examining school reopenings using a difference-in-differences approach, Hansen, Sabia, and Schaller (2022) demonstrate that K-12 reopenings are correlated with significant increases in employment and hours worked by married women. [Hansen et al., 2022] The inconsistency of my results with past research suggests a limitation of my regression specification in correctly estimating the true effect of schooling mode on individual-level labor market outcomes using an averaged state-level measure of schooling mode per month. I confirm the lack of a relationship between averaged shares of time spent in in-person schooling mode in a state and hours worked at the individual level with a graphical analysis. This suggests that an increased level of granularity is needed to examine the effects of schooling mode on individual-level labor market outcomes.

Consequently, I conduct my analysis at the household level restricting my observations to opposite-sex married couple households with both spouses present. I find significant positive results of both in-person and hybrid schooling modes. By shifting from a 0% to 100% in-person schooling mode, I estimate a 27.3 hour increase in total household hours worked and a 22.6 hour increase when this shift is made from hybrid instruction. I observe similar results for total hours actually worked. These results are robust to multiple alternative specifications. Most importantly, however, while the positive effects remain significant when including industry fixed effects, the magnitude of the effects of schooling mode on total hours worked per week substantially decreases by approximately $\frac{2}{3}$. By including this control in the specification, I am able to account for the differences in remote work flexibility offered to individuals. Therefore, it is likely that the true effects of schooling mode on hours worked per week more closely resemble those coefficient estimates where industry FEs are included. This would mean that parents would benefit from an 8.8 hour increase of work per week in a shift from 0% to 100% in-person schooling mode and an increase of 7.1 hours per week in a shift from 0% to 100% hybrid schooling mode.

The most striking observed result is found in examining heterogeneity in household-level treatment. I find that when decomposing parents by subgroups depending on their children's ages, the highest effect of in-person schooling mode occurs for parents with 14-17-year-olds, while the effects for hybrid schooling mode remain similar for children of all age groups. Generally, younger-aged children pose a greater caretaking burden on their parents. However, the unique circumstances surrounding the pandemic may actually explain these results. Given that older children tend to be more autonomous, they pose a greater risk of catching COVID-19 and bringing it back to the household when left without supervision. Hence, the pandemic may have forced parents to take on the burden of childcare for older children. However, further analysis would be needed for such conclusions.

My findings contribute to literature on the effects of school closures on women's labor market outcomes. Most studies to date focus on the effect of school closures ([Heggeness, 2020] & [Garcia and Cowan, 2022]) rather than school reopenings. However, Hansen, Sabia, and Schaller (2022) estimate that school reopenings positively affect the labor market outcomes of married women from the CPS with school-aged children at the extensive and

intensive margins. Similarly to my results, they find that in-person schooling has the largest effect on mothers with older children. [Hansen et al., 2022]

This paper adds to the research produced on the pandemic-driven gender gap in labor market outcomes documented by Stantcheva (2022), Goldin (2022), and Alon et al. (2020a). Alon et. al (2020a) show that women provide 60% of childcare hours among couples with children that work full time—a phenomenon they predict would persist throughout the pandemic. [Alon et al., 2020a] Goldin outlines the extensive impact of the recession triggered by the pandemic on female employment and labor market attachment, highlighting that labor force participation of women fell. [Goldin, 2022] According to research conducted by Collins et al. (2021a), mothers of young children experienced a four to five times higher decrease in work hours than fathers. [Collins et al., 2021a]

My analysis most closely resembles that conducted using data from the CSDH. Halloran et al. (2022), the creators of the data, study the impact of district-level schooling mode on student test scores providing evidence that districts with full in-person instruction significantly decreased their declines in pass rates. [Halloran et al., 2022] Furthermore, Cook (2022) found that increases in district virtual schooling mode in Ohio led to increases in women's uptake of unemployment insurance claims. [Cook, 2022]

While my results do not unilaterally contribute to the literature supporting the importance of in-person instruction on the improvement of the gender disparities that have arisen as a result of the COVID-19 pandemic, they do confirm that in-person schooling is important to improving household labor market outcomes. My analysis is unable to separate the incidence of the impact of schooling mode on mothers as opposed to fathers in opposite-sex married couple households. This would prove to be a valuable further extension to the analysis conducted in this paper. From a policy standpoint, my results indicate the importance of in-person instruction for household economic well-being and should caution against shifting away from in-person teaching in the future.

2 Data

My analysis uses multiple sources of data: 1) Basic Monthly Current Population Survey (CPS) data; 2) school- and district-level learning model data from the 2020-21 school year;

3) additional state-level data used for controls. I explain the data sources in the following subsections.

2.1 Basic Monthly Current Population Survey Data

The labor market outcomes analyzed here come from the Basic Monthly Current Population Survey (CPS). The data used spans years 2019 through 2022 from the public use files published by the National Bureau of Economic Research. [NBER, 2022] The CPS is conducted on a monthly basis and is the U.S. Government's method of tracking unemployment and labor force participation.

The CPS provides the outcome variables of interest for analysis including individuals' total hours worked, hours taken off of work (for any reason), and total hours actually worked—a variable of specific interest during the pandemic, since hours of remote work do not always equal numbers of hours worked in-person. I drop observations of these outcome variables which are negative (left blank by respondents). For those who indicate high variation in hours, but report typical work of at least 35 hours per week, I standardize total hours worked to 35 hours. This grants the most generous minimum number of hours worked possible.

The CPS also collects demographic data. I use demographic variables such as race, education level, and gender, which have explanatory power for labor market outcomes, as controls. I use the gender variables (limited to male and female) to distinguish between how schooling mode has differently affected the total hours worked by men (fathers) and women (mothers) during 2020-21 school year.

To perform household-level analysis, I collapse the CPS individual-level data to opposite-sex married couple households where both spouses are present. I classify these households into parent and non-parent types and include the same demographic variable controls for both individuals in the household as are present in the individual-level analysis.

2.2 Schooling Mode Data

I merged the basic monthly CPS data with schooling mode data from the COVID-19 School Data Hub. [COVID-19 School Data Hub] The COVID-19 School Data Hub (CSDH)

is a public use database compiled by Professor Emily Oster and the CSDH research team with the aim of increasing understanding about how COVID-19 affected schooling across the United States. The CSDH gathers data from state agencies about learning models adopted in K-12 schools and districts over the 2020-21 school year. There were three possible classification of learning models: 1) "in-person" (all or most students had access to traditional, in-person schooling 5 days/week); 2) "virtual" (all or most students received online schooling 5 days/week); 3) "hybrid" (all other learning models). Importantly, the CSDH database reports students' access to a given learning model, which does not guarantee that the students in a given school or district *actually* learned through said approach. Therefore, this analysis estimates the impact of access to a given schooling mode. [Halloran et al., 2022]

SEAs could chose to report their data to the CSDH at the school or district level. I include both reporting types, but restrict my analysis only to those states which reported on a monthly, bi-weekly, or weekly basis due to the lack of variation in share of time spent in different schooling modes for longer reporting timeframes.

I use the CSDH data to create state-specific variables for each learning model each month. I construct this variable by calculating the amount of time that was spent in a given schooling mode in each month and classify each observation to a given month for each unit (individual/household) of observation. I then average these results across the entire month for each state to obtain state-specific monthly observations on schooling mode.

In the U.S., different states start and end their school years at different times. Therefore, I restrict my analysis to months that have the majority of the states reporting data and those that would not be typically affected by vacation periods. To this end, I restrict my analysis to the October 2020 through May 2021 period, while dropping December observations due to variations in length of Christmas and New Year's holidays. I include these months to conduct a robustness check on both individual- and household-level regressions.

2.3 Additional Data Sources

In order to control for state demographic characteristics associated with the COVID-19 pandemic, I merge monthly CPS and schooling mode data with state data on COVID-19 cases and election data from the 2020 Presidential Election. The COVID-19 case count data comes from the U.S. Centers for Disease Control and Prevention (CDC) COVID Data Tracker. [CDC] I use the monthly new case counts for each state in each month and present them in my data as a Covid Case Rate per 100,000 people. Since New York state and New York City report their cases separately to the federal government, I combine these observations before creating a Covid Case Rate variable for NY State. On the other hand, the 2020 Presidential Election Data comes from the MIT Election Data and Science Lab. [MIT Election Data and Science Lab, 2017] I construct the 2020 Republican Vote Share for each state and merge it with my dataset to create a demographic control that could affect the both labor market outcomes and pandemic response. I test the relevance of including this variable in my individual and household level regressions later in this section.

2.4 Summary Statistics

This section presents the summary statistics of variables used in this paper. I construct two tables with summary statistics to help compare the labor market outcomes of individuals across the United States at the beginning and end of the 2020-21 school year.

Table 1 presents the descriptive statistics of weekly labor market outcomes in October 2020. Summary statistics are reported for individuals in the 38 states used in the analysis and are presented separately for men, women, and overall. Table 2 similarly presents the descriptive statistics of weekly labor market outcomes in May 2021. The results of these tables show that men in the states analyzed generally work longer hours per week at their main job and in total as compared to women. On average, they work 4.002 total hours longer than women in October 2020 and 3.668 total hours longer than women in May 2021. This is consistent in showing that as the pandemic progressed and in-person schooling increased thereby decreasing the home production burden, female number of hours worked increased. A similar trend can be observed when looking at total hours actually worked.

On average, men actually worked 4.22 hours per week more than females in October 2020, which decreased to 3.76 hours more in May 2021. Since similar patterns are observed for hours worked and hours actually worked both for the main job and in total, the rest of this paper is concerned with total hours worked and total hours actually worked as outcome variables.

Table 1: Descriptive Statistics of Weekly Labor Market Outcomes in October 2020

	n	mean	sd	median	min	max
Men						
Hours Worked (Main Job)	18167	40.537	10.814	40	0	99
Total Hours Worked	18167	41.123	11.349	40	0	140
Hours Taken Off	1856	12.698	9.929	8	0	84
Hours Actually Worked (Main Job)	17693	39.869	12.139	40	0	99
Total Hours Actually Worked	17693	40.456	12.725	40	1	170
Women						
Hours Worked (Main Job)	16706	36.434	10.927	40	0	99
Total Hours Worked	16706	37.121	11.452	40	0	149
Hours Taken Off	1895	11.755	8.455	8	0	72
Hours Actually Worked (Main Job)	16105	35.639	11.844	40	0	99
Total Hours Actually Worked	16105	36.326	12.339	40	1	137
Total						
Hours Worked (Main Job)	34873	38.597	11.059	40	0	99
Total Hours Worked	34873	39.231	11.571	40	0	149
Hours Taken Off	3751	12.230	9.238	8	0	84
Hours Actually Worked (Main Job)	33798	37.882	12.185	40	0	99
Total Hours Actually Worked	33798	38.516	12.713	40	1	170

Notes: This table presents the summary statistics on labor market outcomes in October 2020 of men, women, and both genders overall in the 38 states selected for the analysis. The 38 states with school learning mode data are: AK, AL, AR, AZ, CA, CO, CT, GA, ID, IL, IN, KS, KY, LA, MA, MD, ME, MI, MN, MO, MS, ND, NE, NH, NJ, NM, NV, NY, OH, OR, RI, SC, UT, VT, WA, WI, WV, WY. The data summarized in this table comes from the Basic Monthly CPS. Observations are weighted by final person weights from the CPS dataset.

Tables 3 and 4 present the summary statistics of weekly labor market variables at the household level in October 2020 and May 2021 respectively. These tables provide the summary statistics for the household-level analysis conducted in this paper. The labor market outcomes presented in these tables are the summed hours of both spouses. I report these summary statistics all opposite-sex married couple households by parent or non-parent type. The results show that on average the number of total hours worked for parent and non-parent households did not change much between October 2020 and May 2021. Furthermore, there is a smaller difference in total hours worked and total

Table 2: Descriptive Statistics of Weekly Labor Market Outcomes in May 2021

	n	mean	sd	median	min	max
Men						
Hours Worked (Main Job)	17432	40.222	10.337	40	0	99
Total Hours Worked	17432	40.910	11.003	40	0	159
Hours Taken Off	1261	12.954	10.533	8	0	80
Hours Actually Worked (Main Job)	16970	39.912	11.402	40	0	99
Total Hours Actually Worked	16970	40.573	11.990	40	1	159
Women						
Hours Worked (Main Job)	16069	36.567	10.602	40	0	99
Total Hours Worked	16069	37.242	11.100	40	0	119
Hours Taken Off	1219	12.201	9.838	8	0	60
Hours Actually Worked (Main Job)	15527	36.155	11.372	40	0	99
Total Hours Actually Worked	15527	36.814	11.822	40	1	119
Total						
Hours Worked (Main Job)	33501	38.498	10.621	40	0	99
Total Hours Worked	33501	39.179	11.199	40	0	159
Hours Taken Off	2480	12.591	10.209	8	0	80
Hours Actually Worked (Main Job)	32497	38.147	11.541	40	0	99
Total Hours Actually Worked	32497	38.807	12.058	40	1	159

Notes: This table presents the summary statistics on labor market outcomes in May 2021 of men, women, and both genders overall in the 38 states selected for the analysis. The 38 states with school learning mode data are: AK, AL, AR, AZ, CA, CO, CT, GA, ID, IL, IN, KS, KY, LA, MA, MD, ME, MI, MN, MO, MS, ND, NE, NH, NJ, NM, NV, NY, OH, OR, RI, SC, UT, VT, WA, WI, WV, WY. The data summarized in this table comes from the Basic Monthly CPS. Observations are weighted by final person weights from the CPS dataset.

hours actually worked between parent and non-parent households in both months than the difference between average male and female hours worked presented in Table 1 and Table 2 above. This appears to indicate that in opposite-sex married households, one partner could have been compensating for the losses in hours of another by working more during pandemic months.

Table 5 presents the descriptive statistics of schooling mode in the 38 states used in the analysis as well as other demographic control variables. The schooling mode variables presented are the share of time of a month in a given school mode. This table shows that on average states spent 0.397 days of the month in-person in October 2020 compared to 0.612 days of the month in May 2021. This is consistent with the idea that schools increased the share of in-person teaching throughout the 2020-21 school year. The additional variables reported in this table include the COVID-19 Case Rate per 100,000

Table 3: Descriptive Statistics of Household Weekly Labor Market Outcomes in October 2020

	n	mean	sd	median	min	max
Non-Parent						
Total Hours Worked	4794	39.988	11.133	40	0	110
Hours Taken Off	475	12.735	9.385	8	0	60
Total Hours Actually Worked	4651	39.270	12.277	40	1	120
Parent						
Total Hours Worked	4163	40.629	10.716	40	0	99
Hours Taken Off	481	12.105	8.156	8	1	60
Total Hours Actually Worked	4025	39.851	12.077	40	1	140
Total						
Total Hours Worked	8957	40.293	10.941	40	0	110
Hours Taken Off	956	12.407	8.768	8	0	60
Total Hours Actually Worked	8676	39.546	12.185	40	1	140

Notes: This table presents the summary statistics on labor market outcomes in October 2020 of opposite-sex married couple households by parent and non-parent household type in the 38 states selected for the analysis. The 38 states with school learning mode data are: AK, AL, AR, AZ, CA, CO, CT, GA, ID, IL, IN, KS, KY, LA, MA, MD, ME, MI, MN, MO, MS, ND, NE, NH, NJ, NM, NV, NY, OH, OR, RI, SC, UT, VT, WA, WI, WV, WY. The data summarized in this table comes from the Basic Monthly CPS. Observations are weighted by the sum of spouses' individual weights from the CPS dataset.

people and the Republican Vote Share from the 2020 Presidential Elections.

A concern for the analysis is that schooling mode was not exogenously assigned to states. If demographic and state-specific variables were correlated with schooling modes and labor market outcomes, then excluding them from the regression could bias the results. In order to understand the correlation between schooling mode and state characteristics, I regressed the share of days in a month spent in-person by state on student school enrollment, COVID-19 Case Rate and Republican Vote Share. Table 6 presents the results of this regression and shows that student school enrollment did not impact the share of in-person learning. Including state fixed effects (FEs) across the 2020-21 school year, I find that higher COVID-19 case rates decrease time spent in-person. When I replace state FEs with Republican Vote Shares from the 2020 Presidential Elections (unique to each state), I find that the negative effect of COVID-19 case rates diminishes, while a higher Republican vote share is correlated with more in-person learning. This is consistent with the fact that states with higher Republican vote shares had less stringent lockdown

Table 4: Descriptive Statistics of Household Weekly Labor Market Outcomes in May 2021

	n	mean	sd	median	min	max
Non-Parent						
Total Hours Worked	4545	39.762	11.049	40	0	110
Hours Taken Off	475	12.735	9.385	8	0	60
Total Hours Actually Worked	4651	39.270	12.277	40	1	120
Parent						
Total Hours Worked	3978	40.782	10.398	40	0	100
Hours Taken Off	481	12.105	8.156	8	1	60
Total Hours Actually Worked	4025	39.851	12.077	40	1	140
Total						
Total Hours Worked	8523	40.251	10.753	40	0	110
Hours Taken Off	956	12.407	8.768	8	0	60
Total Hours Actually Worked	8676	39.546	12.185	40	1	140

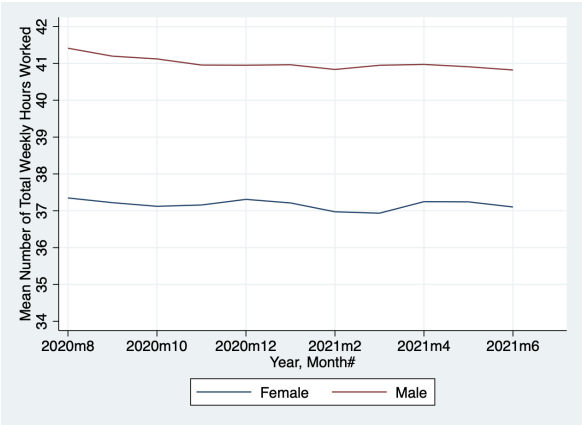
Notes: This table presents the summary statistics on labor market outcomes in May 2021 of opposite-sex married couple households by parent and non-parent household type in the 38 states selected for the analysis. The 38 states with school learning mode data are: AK, AL, AR, AZ, CA, CO, CT, GA, ID, IL, IN, KS, KY, LA, MA, MD, ME, MI, MN, MO, MS, ND, NE, NH, NJ, NM, NV, NY, OH, OR, RI, SC, UT, VT, WA, WI, WV, WY. The data summarized in this table comes from the Basic Monthly CPS. Observations are weighted by the sum of spouses' individual weights from the CPS dataset.

measures and pushed more for in-person schooling. Given that this regression shows a significant negative correlation between COVID-19 case rates and time spent in in-person schooling mode, it confirms the need to include the variable as a control.

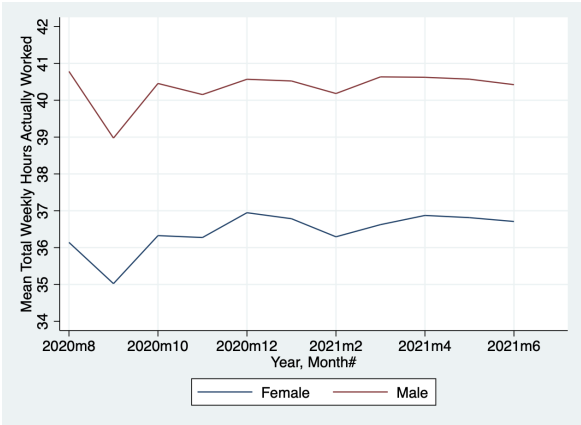
3 Identification Strategy

The aim of this paper is to analyze the impact of schooling mode on labor market outcomes of women as compared to men during the COVID-19 pandemic. The analysis employed is two-fold. The first analysis is carried out at the individual-level separating the effects of schooling mode on labor market outcomes by men and women. The second analysis is carried out at the household-level to try to understand how changes in schooling mode affect household labor market outcomes dividing households into parent and non-parent types. This analysis is an attempt to catch trends that may not present in an analysis at the individual-level.

In both the individual- and household-level analysis, I estimate the impact of schooling

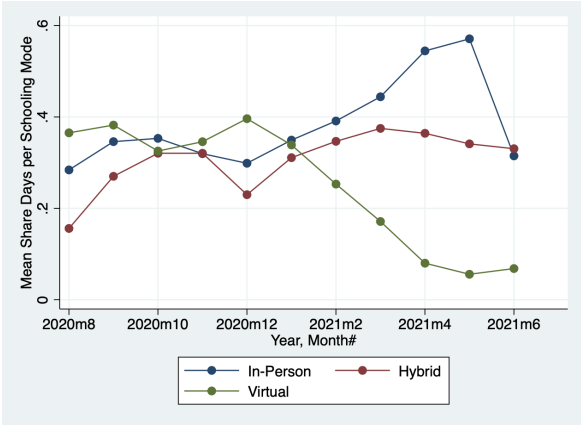


(a) Average Total Number of Weekly Hours Worked by Gender



(b) Average Total Number of Weekly Hours Actually Worked by Gender

(c) Average Number of Weekly Hours Taken Off by Gender



(d) Average Share of Time Spent in Schooling Mode Type

Figure 1: Average of Labor Market Outcomes and Schooling Mode Variables for States Included in Analysis over the 2020-21 School Year

Notes: This figure presents the averages of variables across the 38 states included in the regression analysis at the individual and household level. Each panel presents the average of the variables across the 2020-21 school year: from August 2020 to June 2021. Panel (a) presents the average total number of weekly hours worked by gender. Panel (b) shows the average total number of weekly hours actually worked by gender. Panel (c) shows the number of hours taken off per week by gender. Panel (d) shows the changes in average shares of each schooling mode over the school year.

Table 5: Descriptive Statistics of Schooling Mode and Other Variables

October 2020	n	mean	sd	median	min	max
% In-Person Days	37	.397	.277	.407	0	.917
% Hybrid Days	37	.334	.226	.317	0	.779
% Virtual Days	37	.268	.281	.200	.002	1
Covid Case Rate	38	.350	.295	.262	.004	1.208
Republican Vote Share	38	.496	.108	.489	.304	.695

May 2021	n	mean	sd	median	min	max
% In-Person Days	38	.612	.289	.698	0	1
% Hybrid Days	38	.307	.263	.202	0	1
% Virtual Days	38	.041	.090	.011	0	.497
Covid Case Rate	38	.159	.137	.127	.012	.519
Republican Vote Share	38	.496	.108	.489	.304	.695

Notes: This table presents the summary statistics of in-person, hybrid, and virtual schooling mode across the 38 states with data: AK, AL, AR, AZ, CA, CO, CT, GA, ID, IL, IN, KS, KY, LA, MA, MD, ME, MI, MN, MO, MS, ND, NE, NH, NJ, NM, NV, NY, OH, OR, RI, SC, UT, VT, WA, WI, WV, WY. South Carolina does not report data in 2020 and is excluded from the list of states for schooling mode summary statistics for October 2020. The full list of states is used in constructing the descriptive statistics for May 2021. Additional data on COVID-19 case rates and republican vote shares across the 28 states is included for October 2020 and May 2021.

mode on a labor market outcome variable. The primary treatment variables are state shares of days of a given month spent in either in-person or hybrid schooling mode. The labor market variables of interest are total number of weekly hours worked, total number of weekly hours actually worked, and number of hours taken off/week. I use the following ordinary least squares (OLS) regression in my analysis:

$$y_{ist} = \alpha + \beta(\%In-Person)_{st} + \gamma(\%Hybrid)_{st} + \zeta(CovidRate)_{st} + \eta_s + \delta_t + \Omega X_{ist} + \epsilon_{ist} \quad (1)$$

The variable y_{ist} denotes the labor market outcome for individual i in state s at time t. $(\%In-Person)_{st}$ represents the share of time spent in in-person schooling mode in state s at time t. The term $(\%Hybrid)_{st}$ similarly represents the share of time spent in hybrid schooling mode in state s at time t. The regression also estimates $\zeta(CovidRate)_{st}$, which is a control for the COVID-19 case rate per 100,000 people in state s at time t. The variables η_s state FEs, δ_t time (month of 2020-21 school year) FEs, and a vector of individual-level demographic characteristics X_{ist} are included in the regression. Due to the presence of state and time FEs, the estimates of causal effects of schooling mode on labor

Table 6: Determinants of Schooling Mode

	% In-Person Days of a Month					
Variables	(1)	(2)	(3)	(4)	(5)	(6)
School Enrollment	-0.000	-0.000	-0.000***	-0.000***	-0.000	-0.000***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Covid Case Rate	-0.040	0.059	-0.009	0.001	-0.041***	-0.031**
	(0.156)	(0.109)	(0.015)	(0.011)	(0.016)	(0.015)
Republican Vote Share		1.165***		1.107***		1.055***
		(0.372)		(0.143)		(0.148)
Fixed Effects	none	none	month FEs	month FEs	state FEs	none
Analysis Level	year	year	month	month	month	month
Observations	38	38	370	370	370	370
R-squared	0.110	0.345	0.192	0.335	0.686	0.225

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Notes: This table presents the results of an OLS regression of the share of days spent in in-person schooling mode on relevant pandemic-measure-defining variables. Columns (1) and (2) conduct the regression based on state averages across the entire year. Columns (3) and (4) include month FEs. Columns (5) and (6) present the results of the regression with state FEs. The regressions in columns (3)-(6) are conducted at the monthly level.

market outcomes of an individual are assumed to be stripped of time-specific effects and state-specific effects. Lastly, ϵ_{ist} is an error term assumed to be uncorrelated with other variables not included in this regression that may have an impact on the outcome variable.

The same regression is conducted at the household level. The household-level OLS regression is adjusted as follows:

$$y_{hst} = \alpha + \beta(\%In-Person)_{st} + \gamma(\%Hybrid)_{st} + \zeta(CovidRate)_{st} + \eta_s + \delta_t + \Omega X_{hst} + \epsilon_{hst} \quad (2)$$

The coefficients of interest in both of these regressions are primarily β and γ. They estimate the impact of the share of state in-person and hybrid schooling mode on labor market outcome y_{ist} or y_{hst}, when combined with regression controls. This means my regression specification is attempting to explain the effects of state shares of in-person and hybrid schooling mode in conjunction with state-level and unit-level control variables on

unit-level outcomes (where the unit is either the individual or the household).

I control for COVID-19 case rate per 100,000 people as Table 1 showed that share of time spent in in-person schooling mode can be significantly explained by the state COVID-19 case rate in a given month. Since the level of economic activity in each state during the pandemic was often determined by the degree to which the virus was spreading in the state, the COVID-19 case rate might have had a direct impact on labor market outcomes including the hours individuals worked in given months. Not including this control variable could bias my results by attributing the effect of COVID-19 case rate on labor market outcomes to the coefficients of interest.

I also control for a vector of individual- and household-specific demographic variables. It is well known that educational attainment and race are variables that impact labor market outcomes. Therefore, I include controls for degree of educational attainment achieved, race, as well as marriage status. Marriage status could affect an individual's hours worked as a two-earner household does not depend on one individual's hours worked as much and can sustain declines in hours worked more than a single-earner household.

4 Results

This section discusses the empirical findings of the analysis employed in this paper. It is divided into two main subsections: the first presents the results of the individual-level analysis and the second presents the results of the household-level analysis. Within each subsection, I divide my results into three different categories: 1) primary results; 2) treatment effect heterogeneity; and 3) robustness.

4.1 Individual-Level Analysis

4.1.1 Primary Results

The primary results of the individual-level analysis are presented in Tables 7-9. Tables 7, 8, and 9 show the results of the effect of schooling mode on total weekly hours worked, total weekly hours actually worked, and hours taken off/week respectively. Columns (1)-(3) estimate the impact of the regression for women, while columns (4)-(6) for men. Moving

horizontally across the columns for each gender, I restrict the groups of individuals further to those with and without kids.

I find no significant effects of schooling mode on men's labor market outcomes of total weekly hours worked (Table 7) and total weekly hours actually worked (Table 8). For women, however, I find a significant negative effects of share of time of the month spent in-person on total hours worked for women overall. This effect is highly significant at the 1% level, but diminishes in magnitude and significance when the treatment group is restricted to women with kids. A similar effect is observed in Table 8, where the share of time spent in hybrid schooling mode picks up no effect on total weekly hours actually worked, while there is a significant negative effect observed on women's weekly total hours actually worked of the share of time spent in in-person schooling mode. This effect remains significant when limiting the analysis to women with children.

These observed coefficient estimates are contrary to those found in prior literature. Since a higher share of in-person schooling mode would lead to a smaller childcare burden on women, female hours worked should theoretically increase. Additionally, it appears that the regressions in both Table 7 and 8 capture significant effects of schooling mode on labor market outcomes of women without children—something that should theoretically not occur. This casts doubt about whether state averages of shares of time spent in a given schooling mode can help understand individuals' labor market outcomes. Moreover, there may be general trends in women's individual labor force attachment throughout the pandemic that are unlinked to schooling mode that have affected both parents and childless women and caused them to suffer a decrease in working hours. For instance, Alon et al. (2020) show that women's employment decreased more then men's throughout the pandemic, since lockdowns affected sectors with higher female employment. [Alon et al., 2020b] In analyzing employment and labor force participation changes, Albanesi & Kim (2021) find that women's employment remained largely below pre-pandemic levels in 2020, while women accounted for 70% of the aggregate rise in non-participation with the level rising to 100% of the average as the pandemic progressed from the spring to the fall. [Albanesi and Kim, 2021] If this trend continued into the beginning of 2021, this could likely explain the decrease in number of hours worked by women in a time during which schools were increasingly pushing to reopen.

To confirm the lack of relationship between in-person schooling mode and labor market outcomes suggested by these regression results, I turn to examining graphical evidence. Figure 2 presents the results showing that there is no observed relationship between the share of days spent in person in a given month and hours worked, especially for women.

Table 7: Effect of Schooling Mode on Weekly Total Hours Worked

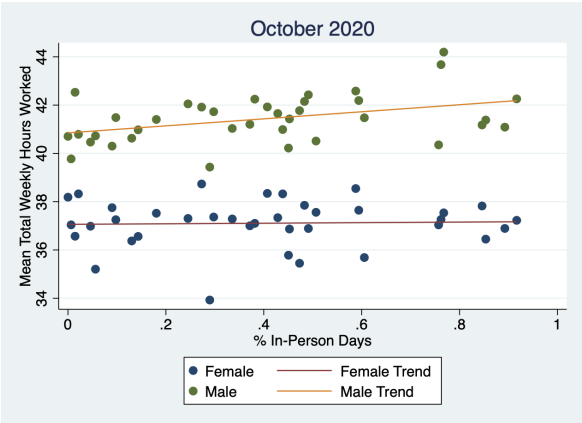
	(1)	(2)	(3)	(4)	(5)	(6)
Gender	women	women	women	men	men	men
Individual Type	overall	no kids	kids	overall	no kids	kids
% In-Person	-1.236***	-1.217**	-1.103*	-0.095	-0.211	0.142
	(0.398)	(0.498)	(0.651)	(0.374)	(0.474)	(0.581)
% Hybrid	0.002	-0.168	0.370	-0.374	-0.631*	0.227
	(0.312)	(0.384)	(0.533)	(0.285)	(0.361)	(0.442)
Covid Case Rate	0.023	0.012	0.035	-0.006	-0.005	-0.004
	(0.035)	(0.043)	(0.061)	(0.032)	(0.041)	(0.051)
HS Degree	5.716***	7.245***	1.455***	4.153***	5.501***	1.600***
	(0.188)	(0.224)	(0.308)	(0.141)	(0.183)	(0.186)
College Degree	2.913***	3.513***	1.972***	1.152***	1.456***	0.545***
	(0.072)	(0.088)	(0.127)	(0.069)	(0.088)	(0.111)
White	-0.709***	-0.718***	-0.557***	0.754***	0.538***	1.305***
	(0.089)	(0.110)	(0.150)	(0.087)	(0.110)	(0.138)
Married	0.530***	1.164***	-0.887***	3.135***	2.589***	1.252***
	(0.071)	(0.087)	(0.133)	(0.067)	(0.083)	(0.160)
State FEs	yes	yes	yes	yes	yes	yes
Month FEs	yes	yes	yes	yes	yes	yes
Observations	125,728	84,582	41,146	136,398	93,511	42,887
R-squared	0.044	0.065	0.019	0.045	0.044	0.021

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

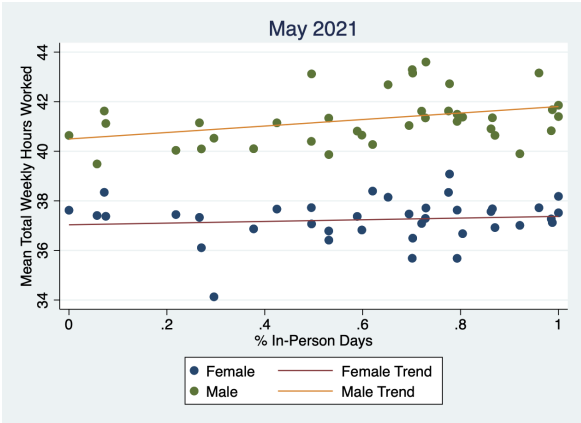
Notes: This table presents the effects of schooling mode on women's and men's total weekly hours worked by parent type for the analysis period of October 2020 to May 2021 (excluding December 2020) for the set of 38 states with schooling mode data available. All regressions are weighted by the final person weights from the CPS. Demographic controls are included for all regressions and coefficient estimates are shown here. State and month fixed effects (FEs) are included in all regressions.

4.1.2 Treatment Effect Heterogeneity

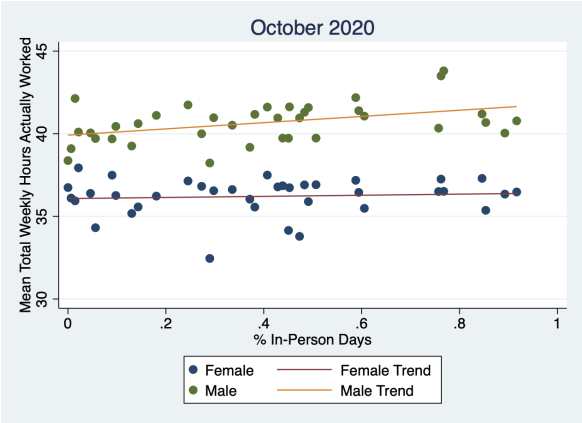
Next, I turn to refining the treatment groups in order to check whether there are effects of schooling mode not captured in analysis at the individual level. To estimate this treatment



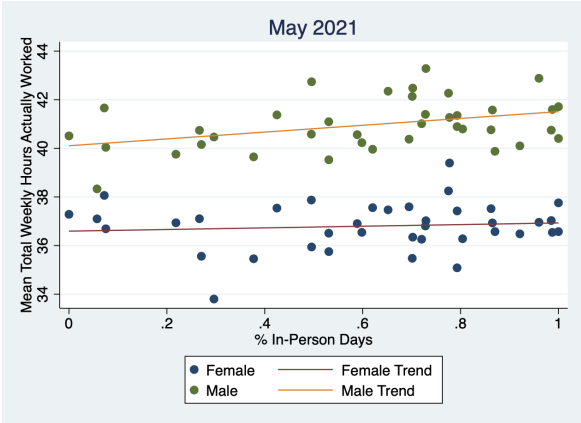
(a) Average Total Weekly Hours Worked by Gender in October 2020



(b) Average Total Weekly Hours Worked by Gender in May 2021



(c) Average Total Weekly Hours Actually Worked by Gender in October 2010



(d) Average Total Weekly Hours Actually Worked by Gender in May 2021

Figure 2: Relationship Between Average of Hours Worked and Share of Month of In-Person Schooling Mode by Gender in the 2020-21 School Year

Notes: This figure presents the relationships between shares of the month spend in in-person schooling mode and labor market outcomes: total weekly hours worked and total weekly hours actually worked. Each panel decomposes the relationship in the given month by gender. Means of the labor market outcomes are constructed from individuals' observations for each states included in the analysis in the given month. October 2020 represents the month of the beginning of the analysis for the 2020-21 school year, while May 2021 represents the month concluding the analysis for the school year. The shares of time spent in in-person schooling mode represent the percent of days of a given month spent in in-person schooling mode. All scatter plots are weighted by final person weights from the CPS. The trendline in each panel shows the line of best fit using OLS.

Table 8: Effect of Schooling Mode on Weekly Total Hours Actually Worked

	(1)	(2)	(3)	(4)	(5)	(6)
Gender	women	women	women	men	men	men
Individual Type	overall	no kids	kids	overall	no kids	kids
% In-Person	-1.477***	-1.309**	-1.682**	0.142	0.182	0.044
	(0.435)	(0.541)	(0.724)	(0.423)	(0.528)	(0.687)
% Hybrid	-0.126	-0.222	0.090	-0.447	-0.724*	0.157
	(0.343)	(0.421)	(0.589)	(0.323)	(0.403)	(0.528)
Covid Case Rate	0.004	0.007	-0.014	-0.011	-0.014	0.002
	(0.039)	(0.048)	(0.069)	(0.036)	(0.046)	(0.058)
HS Degree	5.546***	7.024***	1.429***	4.382***	5.634***	2.036***
	(0.198)	(0.235)	(0.338)	(0.153)	(0.195)	(0.220)
College Degree	2.935***	3.549***	1.924***	1.067***	1.417***	0.363***
	(0.080)	(0.098)	(0.142)	(0.078)	(0.098)	(0.130)
White	-0.688***	-0.662***	-0.622***	0.727***	0.512***	1.258***
	(0.099)	(0.121)	(0.168)	(0.097)	(0.122)	(0.159)
Married	0.529***	1.105***	-0.736***	3.006***	2.450***	1.122***
State FEs	yes	yes	yes	yes	yes	yes
Month FEs	yes	yes	yes	yes	yes	yes
Observations	121,210	81,758	39,452	132,411	90,628	41,783
R-squared	0.039	0.057	0.016	0.038	0.039	0.018

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Notes: This table presents the effects of schooling mode on women's and men's total weekly hours actually worked by parent type for the analysis period of October 2020 to May 2021 (excluding December 2020) for the set of 38 states with schooling mode data available. All regressions are weighted by the final person weights from the CPS. Demographic controls are included for all regressions and coefficient estimates are shown here. State and month fixed effects (FEs) are included in all regressions.

effect heterogeneity, I restrict my sample to parents. Tables A2-A3 present results for total weekly hours worked and total weekly hours actually worked based on: 1) kids' ages; 2) parent education level; and 3) year of analysis (see: Appendix).

The results of the analysis of schooling mode on total weekly hours worked (Panel A) and total weekly hours actually worked (Panel B) by kids' ages are presented in Table A1 (see: Appendix). This regression also captures no significant effect of schooling mode on men's labor market outcomes and a significant negative effect for both labor market outcomes variables at the 1% significance level for women with children between the ages of 0-13. For women with children between the ages of 6-13, the regression also estimates

Table 9: Effect of Schooling Mode on Hours Taken Off / Week

	(1)	(2)	(3)	(4)	(5)	(6)
Gender	women	women	women	men	men	men
Individual Type	overall	no kids	kids	overall	no kids	kids
% In-Person	-1.566	-2.155	-0.173	-1.224	0.005	-2.909
	(1.155)	(1.441)	(1.984)	(1.165)	(1.453)	(1.932)
% Hybrid	-0.890	-0.719	-1.162	0.487	0.422	1.098
	(0.780)	(0.968)	(1.338)	(0.840)	(0.999)	(1.550)
Covid Case Rate	0.291**	0.297*	0.277	0.145	0.186	0.069
	(0.121)	(0.154)	(0.195)	(0.118)	(0.132)	(0.243)
HS Degree	-0.303	-0.102	-0.730	0.290	0.324	0.345
	(0.443)	(0.589)	(0.681)	(0.343)	(0.448)	(0.546)
College Degree	-0.431**	-0.573**	-0.123	-0.916***	-0.738***	-1.282***
	(0.179)	(0.223)	(0.325)	(0.194)	(0.239)	(0.334)
White	-0.568**	-0.415	-0.905**	0.185	0.226	0.035
	(0.231)	(0.284)	(0.393)	(0.248)	(0.303)	(0.433)
Married	-0.052	-0.004	-0.130	0.793***	0.863***	0.648
	(0.173)	(0.215)	(0.345)	(0.187)	(0.233)	(0.493)
State FEs	yes	yes	yes	yes	yes	yes
Month FEs	yes	yes	yes	yes	yes	yes
Observations	13,537	8,846	4,691	14,008	9,311	4,697
R-squared	0.015	0.015	0.025	0.027	0.024	0.052

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Notes: This table presents the effects of schooling mode on women's and men's total hours taken off per week by parent type for the analysis period of October 2020 to May 2021 (excluding December 2020) for the set of 38 states with schooling mode data available. All regressions are weighted by the final person weights from the CPS. Demographic controls are included for all regressions and coefficient estimates are shown here. State and month fixed effects (FEs) are included in all regressions.

an effect of -2.026 on total hours actually worked at the 5% significance level.

Table A2 presents the results of schooling mode on total weekly hours worked and total weekly hours actually worked by gender and parent education level. In most regressions, the coefficient estimates of in-person and hybrid schooling mode remain insignificant. For total weekly hours actually worked, there are significant negative coefficient estimates of the effect of in-person schooling mode for women with a high school degree and women with a college degree. Since in-person schooling mode is unlikely to have a negative effect on labor market outcomes especially those of women with higher educational attainment,

perhaps the pandemic caused these women to refocus their priorities and decrease hours actually worked. Furthermore, the remaining pandemic workplace flexibility measures may have decreased the number of hours women actually work despite their children attending school in-person. It would be effective to test whether workplace flexibility measures decreased hours that were actually worked during the pandemic.

4.1.3 Robustness

Considering the negative effects of in-person and hybrid schooling mode observed at the individual-level analysis, I conduct several robustness exercises to better understand the significance and validity of my results. Tables A4-A6 present these results for total weekly hours worked and total weekly hours actually worked (see: Appendix).

Table A4 presents the results of the robustness of the results to including additional months. For the regressions in this table, I include the previously excluded months: August, September, December 2020 and June 2021. The negative coefficient estimates for the effect of in-person schooling mode on hours worked decrease both in magnitude and significance.

Similar results are observed in Table A5 which presents the robustness of the results to dropping observations with allocation flags for outcome variables. Given that the CPS is a survey, which may not be completely filled out by participants, sometimes the Census Bureau allocates a value for an individual's response. These allocations typically occur for variables where the individual has a value of "don't know" or "refused" and are noted with an allocation flag. Dropping observations that include an allocation flag for the labor market outcome variable restricts the sample of analysis to only those observations where the survey responses are present and have been unedited.

Lastly, Table A6 presents the results of robustness to including industry fixed effects. A concern with the previous specification could be that individuals in different types of jobs would have different remote work possibility and therefore would experience different patterns of labor market outcomes. Including major industry fixed effects would effectively control for this variation. The results of Table A6 present similar results to those of the other robustness specifications estimating no significant effect of schooling mode on father hours worked and negative effects for mothers.

4.2 Household-Level Analysis

4.2.1 Primary Results

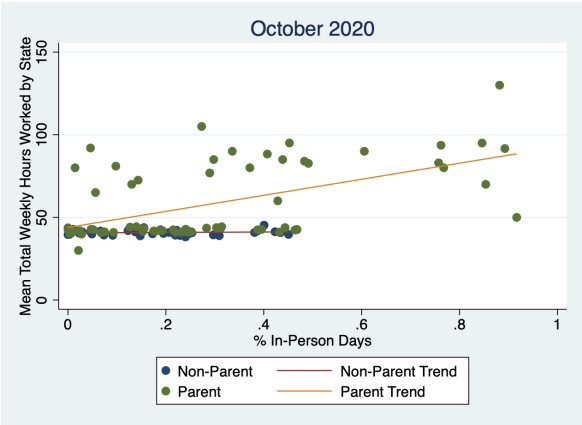
The primary results of the household-level analysis are presented in Table 10. Columns (1)-(2) present results of the regression on total hours worked, columns (3)-(4) for total hours actually worked, and columns (5)-(6) for hours taken off per week. The regression on each labor market outcome is performed for opposite-sex married households (where the spouse is present) overall and restricting the sample only to parent households.

Table 10 shows a positive significant effect of both in-person schooling mode and hybrid schooling mode on total weekly household hours worked. I observe a 24.348 hour increase of total hours worked due to in-person schooling mode and a 20.793 hour increase due to hybrid schooling mode. These results are augmented when the household type is restricted only to parent households, which is consistent with the causal mechanism. Similar effects are observed for total household hours actually worked with 25.175 hour increase and a 21.383 hour increase in hours actually worked per week due to a shift from 0% to 100% in-person schooling mode and hybrid schooling mode respectively. There is a positive effect of in-person and hybrid schooling modes on hours taken off by the household, but it is insignificant at the 5% level. These results appear to indicate that my regression specification is more adept at explaining the changes of household labor market outcomes in response to variations in schooling mode rather than explaining these outcomes at the individual-level.

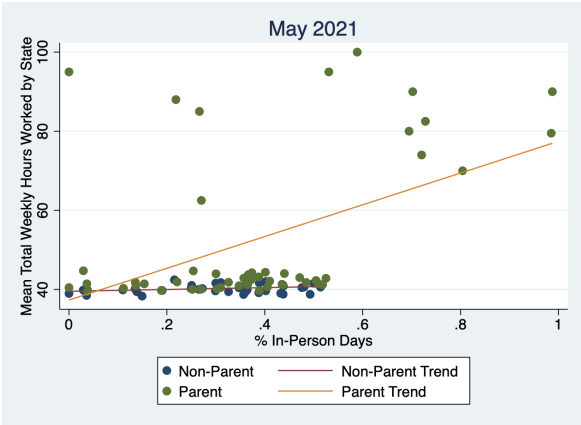
Figure 3 graphically confirms the relationship observed in the regression results of Table 10. Non-parent households appear to be unaffected in labor market outcomes by the share of days of a given month spent in in-person schooling mode across the entire school year. Importantly, however, there is a positive relationship between in-person schooling mode and average weekly hours worked for parental households across the 2020-21 school year, which verifies the existence of the causal mechanism on which this analysis is built.

4.2.2 Treatment Effect Heterogeneity

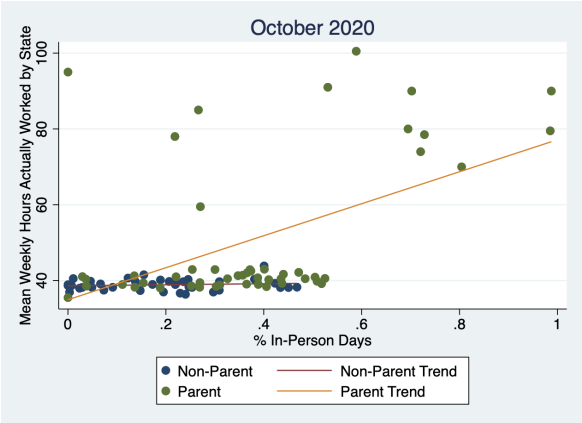
To understand how different types of parental households are affected by share of the month spent in in-person schooling mode, I conduct an analysis of treatment effect heterogeneity



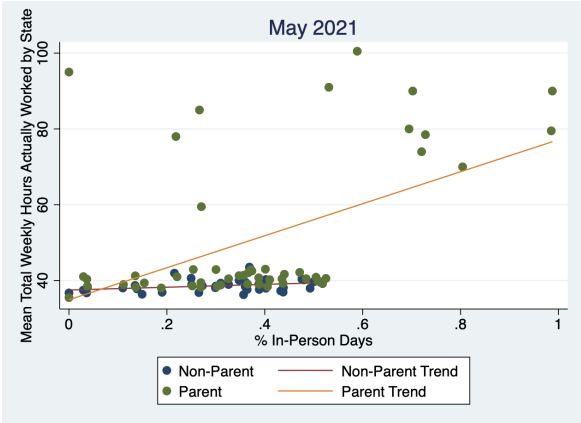
(a) Average Total Weekly Hours Worked by Household Type in October 2020



(b) Average Total Weekly Hours Worked by Household Type in May 2021



(c) Average Total Weekly Hours Actually Worked by Household Type in October 2010



(d) Average Total Weekly Hours Actually Worked by Houshold Type in May 2021

Figure 3: Relationship Between Average Hours Worked and In-Person Schooling Mode at the Household Level in the 2020-21 School Year

Notes: This figure presents the relationships between shares of the month spent in in-person schooling mode and labor market outcomes at the household level: total weekly hours worked and total weekly hours actually worked. Households included in this analysis are opposite-sex spouses. Each panel decomposes the relationship in the given month by household type: parent or non-parent. Means of the labor market outcomes are constructed from household observations for each state included in the analysis for the given month. October 2020 represents the month of the beginning of the analysis for the 2020-21 school year, while May 2021 represents the month concluding the analysis. The shares of time spent in in-person schooling mode represent the percent of days of a given month spent in in-person schooling mode.

Table 10: The Effects of Schooling Mode on Weekly Household Labor Market Outcomes - Household Level

	(1)	(2)	(3)	(4)	(5)	(6)
Labor Market Outcome	**Total Hours Worked**		**Hours Actually Worked**		**Hours Off**	
Household Type	overall	parent	overall	parent	overall	parent
% In-Person	24.348***	27.341***	22.082***	25.175***	3.506*	2.948
	(0.852)	(1.177)	(0.901)	(1.257)	(1.987)	(2.676)
% Hybrid	20.793***	22.582***	19.440***	21.383***	1.994	-0.178
	(0.726)	(1.015)	(0.776)	(1.084)	(1.552)	(2.105)
Covid Case Rate	1.261***	1.415***	1.187***	1.329***	0.765**	0.975*
	(0.119)	(0.167)	(0.131)	(0.186)	(0.370)	(0.504)
State FEs	yes	yes	yes	yes	yes	yes
Month FEs	yes	yes	yes	yes	yes	yes
Observations	67,490	31,862	65,370	30,866	7,121	3,354
R-squared	0.177	0.254	0.147	0.210	0.042	0.069

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Notes: This table presents the effects of schooling mode on total household labor market outcomes by household type for the analysis period of October 2020 to May 2021 (excluding December 2020) for the set of 38 states with schooling mode data available. The household labor market outcomes are limited to opposite-sex married couple households. Columns (1) and (2) present the effects of schooling mode on total weekly household hours worked, columns (3) and (4) on total weekly hours actually worked, and columns (5) and (6) on total household hours taken off per week. All regressions are weighted by the sum of spouses' final person weights from the CPS. Demographic controls are included for all regressions, but coefficient estimates are omitted here. State and month fixed effects (FEs) are included in all regressions.

at the household-level.

Table 11 confirms the positive effect of in-person schooling mode on parent hours worked for each range of children's ages. The coefficient estimates are all significant at the 1% level. The result is, rather surprisingly, most pronounced for households with kids aged 14-17 given that parents with younger children have a larger childcare burden when kids are staying home. However, parental caretaking burden on older children may have possibly increased as a result of the pandemic as parents try to shield their children from infection. The autonomy of older children may in some ways actually pose a hindrance as parents need to watch over the actions of their older children outside the household more closely. The overall effect of in-person schooling on children aged 14-17 is an increase of

30.880 hours worked, for kids aged 3-5 it is 28.406 hours worked, and for kids aged 6-13 it is 26.803 hours worked.

Table 11: Analyzing Heterogeneity in Effects of Schooling Mode on Household Parent Total Hours Worked by Kids' Ages - Household Level

	(1)	(2)	(3)	(4)
Kids' Ages	0-13 years	3-5 years	6-13 years	14-17 years
% In-Person	24.876***	28.406***	26.803***	30.880***
	(1.404)	(2.301)	(1.623)	(1.796)
% Hybrid	21.257***	24.138***	23.098***	24.585***
	(1.141)	(1.943)	(1.338)	(1.663)
Covid Case Rate	1.194***	1.783***	1.436***	1.501***
	(0.171)	(0.308)	(0.205)	(0.278)
State FEs	yes	yes	yes	yes
Month FEs	yes	yes	yes	yes
Observations	25,827	8,758	18,744	11,758
R-squared	0.225	0.295	0.251	0.268

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Notes: Table 11 replicates the results of Table 10 for total household hours worked at the household-level decomposed by heterogeneity treatment effects. The household labor market outcomes are limited to opposite-sex married couple households. Heterogeneity treatment effects are examined in terms of parent type divided by kids' ages. Table 11 only presents results for one labor market outcome (the heterogeneous effects of schooling mode on parental type by kids' ages are not presented here for total weekly hours actually worked or for total hours taken off per week). All regressions are weighted by the sum of spouses' final person weights from the CPS. Demographic controls are included for all regressions, but coefficient estimates are omitted here. State and month fixed effects (FEs) are included in all regressions.

4.2.3 Robustness

Considering the difference in coefficient estimates proposed by my analyses, I conduct several robustness checks to understand the significance and validity of my household-level results.

Table A7 shows that results for total hours worked and total hours actually worked are robust to including additional months in my analysis. While the results are significant, the magnitude decreases. This is probably due higher variability in hours worked by parents during summer and winter months during which children have time off from school and may require more parental attention. Also, school vacations often overlap with time

off taken off work. Interestingly, however, this robustness exercise presents no significant effects of schooling mode on hours taken off per week.

Next, I turn to examining the robustness to dropping observations with allocation flags. Table A8 shows that my results remain robust, with estimated coefficients remaining significant and retaining the magnitude observed in the primary results.

My final robustness exercise entails including industry fixed effects. As shown in Table 12, the significant positive effects of both in-person and hybrid schooling modes on household hours worked remain significant to this specification, but substantially decrease in magnitude. As was explained above, including industry FEs controls for different remote working possibilities during the pandemic in major industries classified by the CPS. The estimated coefficients in the regression decrease by approximately $\frac{2}{3}$ across all columns for both labor market outcomes (hours taken off are omitted due to their demonstrated lack of responsiveness to changes in schooling mode). Consequently, it is likely that the true effects of in-person and hybrid schooling modes more closely represent those observed in this regression specification. Nevertheless, the potential of in-person schooling's positive impact on household-level labor market outcomes cannot be ignored. An approximately eight hour increase in household hours worked per week translates to 32 extra hours worked by the household in the month, which has the power to significantly increase earnings, especially if individuals are employed on a wage-basis.

Table 12: Robustness to Including Industry FEs — Household Level

	(1)	(2)	(3)	(4)
	Total Hours Worked		**Hours Actually Worked**	
Level	overall	parent	overall	parent
% In-Person	7.781***	8.842***	6.367***	7.702***
	(0.887)	(1.268)	(0.941)	(1.348)
% Hybrid	7.040***	7.142***	6.217***	6.625***
	(0.753)	(1.095)	(0.806)	(1.152)
Covid Case Rate	0.509***	0.527***	0.438***	0.464***
	(0.105)	(0.145)	(0.116)	(0.164)
State FEs	yes	yes	yes	yes
yes	yes			
Month FEs	yes	yes	yes	yes
yes	yes			
Observations	66,708	31,521	64,483	30,469
R-squared	0.257	0.346	0.213	0.288

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Notes: Table 14 replicates the results of Table 10 at the household level extending the analysis to industry FEs for each spouse using major industry codes from the CPS. All regressions are weighted by the sum of spouses' final person weights from the CPS. Demographic controls are included for all regressions, but coefficient estimates are omitted here. State and month fixed effects (FEs) are included in all regressions.

5 Conclusion

This paper analyzes the effect of schooling mode on parental labor market outcomes at the individual and household levels. Overall, I find insignificant effects of in-person and hybrid schooling mode at the individual-level. However, I find significant increases in labor market outcomes as a result of shifting away from virtual instruction to in-person and hybrid schooling modes at the household-level. These effects were most concentrated among parents of older children. Controlling for industry fixed effects substantially decreases the magnitude of these results suggesting that the true effects of schooling mode may be lower than those reported in the primary results.

While this study does not confirm that in-person schooling improves mothers' labor market outcomes, it does point to such results at the household level for opposite-sex married parent households. An extension to this analysis conducted at a more granular

level would be necessary to identify a relationship between mothers' labor market outcomes and schooling mode. While there exist other fruitful avenues for research on the effects of school closures, further future studies could also be conducted on the effects of remaining pandemic workplace flexibilities on female labor market outcomes. Since women have been found to appreciate workplace flexibility more than men, such policies could lead to improvements in their long-term employment outcomes. [Flabbi and Moro, 2012]

The results in this paper should draw the attention of policymakers to focusing on avoiding future school disruptions. While harsh lockdowns hurt the well-being of the economy, they also negatively impact children's educational outcomes and parental labor market outcomes. More importantly, policymakers should consider precautionary measures for the future that could help us avoid the far-reaching negative consequences that school disruptions caused in the spring of 2020 and during the 2020-21 school year.

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7 Appendix

Table A1: Analyzing Heterogeneity in Effects of Schooling Mode on Parent Hours Worked by Kids' Ages

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A				**Total Hours Worked**				
	women	women	women	women	men	men	men	men
Kids' Ages	0-13 years	3-5 years	6-13 years	14-17 years	0-13 years	3-5 years	6-13 years	14-17 years
% In-Person	-1.709**	0.324	-1.227	0.027	-0.346	0.108	-0.493	0.214
	(0.731)	(1.358)	(0.872)	(1.063)	(0.636)	(1.116)	(0.757)	(1.014)
% Hybrid	0.026	0.286	0.781	0.127	-0.328	-0.207	-0.403	1.246
	(0.584)	(1.072)	(0.698)	(0.904)	(0.478)	(0.802)	(0.574)	(0.776)
Observations	32,967	10,254	23,802	15,292	35,478	12,236	24,832	14,923
R-squared	0.023	0.030	0.022	0.017	0.021	0.026	0.025	0.026
Panel B				**Total Hours Actually Worked**				
	women	women	women	women	men	men	men	men
Kids' Ages	0-13 years	3-5 years	6-13 years	14-17 years	0-13 years	3-5 years	6-13 years	14-17 years
% In-Person	-2.400***	-0.598	-2.026**	-0.633	-0.504	0.865	-0.489	0.112
	(0.815)	(1.497)	(0.963)	(1.150)	(0.768)	(1.321)	(0.937)	(1.153)
% Hybrid	-0.408	0.022	0.473	0.209	-0.216	0.305	-0.449	0.574
	(0.647)	(1.154)	(0.767)	(0.994)	(0.576)	(0.943)	(0.703)	(0.923)
Observations	31,480	9,806	22,945	14,843	34,562	11,938	24,221	14,550
R-squared	0.019	0.024	0.019	0.017	0.019	0.022	0.021	0.024

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Notes: Table A1 presents the results of heterogeneity treatment effects for parents at the individual-level analysis decomposing the effect on different parental groups by kids' ages. Panel A reproduces Table 7 and Panel B reproduces Table 8 for different parental groups. All regressions are weighted by final person weights from the CPS. Demographic controls are included for all regressions (including controls for COVID-19 case rates), but coefficient estimates are omitted here. State and month fixed effects (FEs) are included in all regressions.

Table A2: Analyzing Heterogeneity in Effects of Schooling Mode on Parent Hours Worked by Parent Education Level

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A			**Total Hours Worked**			
	women	women	women	men	men	men
Education Level	dropout	HS degree	college degree	dropout	HS degree	college degree
% In-Person	-0.274	-1.220*	-1.311	2.103	0.030	0.858
	(3.156)	(0.666)	(0.904)	(2.099)	(0.604)	(0.875)
% Hybrid	1.165	0.268	0.665	2.069	0.119	0.539
	(2.430)	(0.546)	(0.758)	(1.457)	(0.467)	(0.683)
Observations	1,926	39,220	19,994	3,379	39,508	18,306
R-squared	0.046	0.011	0.016	0.048	0.017	0.018
Panel B			**Total Hours Actually Worked**			
	women	women	women	men	men	men
Education Level	dropout	HS degree	college degree	dropout	HS degree	college degree
% In-Person	-2.185	-1.692**	-2.536**	1.843	-0.063	0.579
	(3.589)	(0.739)	(1.002)	(2.622)	(0.713)	(0.992)
% Hybrid	2.216	-0.069	0.026	1.195	0.117	0.468
	(2.671)	(0.604)	(0.843)	(1.716)	(0.558)	(0.792)
Observations	1,834	37,618	19,272	3,258	38,525	18,002
R-squared	0.051	0.009	0.016	0.049	0.015	0.017

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Notes: Table A2 presents the results of heterogeneity treatment effects for parents at the individual-level analysis decomposing the effect on different parental groups by parental education level. Panel A reproduces Table 7 and Panel B reproduces Table 8 for different parental groups: the classification of dropouts indicates those parents who did not attain a HS degree, HS Degree indicates the attainment of at least a high school degree and higher, but not a college degree, while college degree indicates educational attainment of at least a college degree and higher. All regressions are weighted by final person weights from the CPS. Demographic controls are included for all regressions (including controls for COVID-19 case rates), but coefficient estimates are omitted here. State and month fixed effects (FEs) are included in all regressions.

Table A3: Analyzing Heterogeneity in Effects of Schooling Mode on Parent Hours Worked by Year

	(1)	(2)	(3)	(4)
Panel A	**Total Hours Worked**			
	women	women	men	men
Year	2020	2021	2020	2021
% In-Person	0.326	-1.017	0.661	0.077
	(2.723)	(0.827)	(2.476)	(0.715)
% Hybrid	-1.302	-0.508	0.011	-0.137
	(1.722)	(0.719)	(1.476)	(0.584)
Covid Case Rate	0.156	-0.055	0.080	-0.031
	(0.315)	(0.068)	(0.264)	(0.056)
Observations	15,154	25,992	15,634	27,253
R-squared	0.025	0.020	0.025	0.021
Panel B	**Total Hours Actually Worked**			
	women	women	men	men
Year	2020	2021	2020	2021
% In-Person	-2.791	-1.293	-2.894	0.546
	(3.259)	(0.905)	(3.061)	(0.827)
% Hybrid	-2.296	-0.684	-0.350	0.270
	(1.954)	(0.793)	(1.876)	(0.688)
Covid Case Rate	0.088	-0.114	0.125	-0.010
	(0.357)	(0.076)	(0.320)	(0.064)
Month FEs	yes	yes	yes	yes
State FEs	yes	yes	yes	yes
Observations	14,485	24,967	15,249	26,534
R-squared	0.021	0.018	0.023	0.018

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Notes: Table A3 presents the results of heterogeneity treatment effects for parents at the individual-level analysis decomposing the effect by parents in different years. Panel A reproduces Table 7 and Panel B reproduces Table 8 for 2020 and 2021 respectively. All regressions are weighted by final person weights from the CPS. Demographic controls are included for all regressions, but coefficient estimates are omitted here. State and month fixed effects (FEs) are included in all regressions.

Table A4: Robustness to Including Additional Months — Individual Level

	(1)	(2)	(3)	(4)
	Total Hours Worked		**Hours Actually Worked**	
Parent Type	mothers	fathers	mothers	fathers
% In-Person	-0.800*	0.152	-0.251	0.834
	(0.478)	(0.413)	(0.642)	(0.576)
% Hybrid	0.211	0.202	-0.378	0.815*
	(0.408)	(0.342)	(0.546)	(0.478)
Covid Case Rate	-0.018	-0.042	-0.086	-0.065
	(0.046)	(0.037)	(0.062)	(0.051)
Month FEs	yes	yes	yes	yes
State FEs	yes	yes	yes	yes
Observations	51,179	53,527	51,179	53,527
R-squared	0.019	0.021	0.014	0.016

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Notes: Table A4 presents the same results as individual-level analysis in Table 7 and Table 8 for mothers and fathers and extends the analysis period to include previously excluded months for the set of 38 states with schooling mode data available. All regressions include summer months (August 2020, September 2020, and June 20201) as well as December 2020. Columns (1) and (2) present the same results as Table 7, while columns (3) and (4) present the same results as Table 8. All regressions are weighted by final person weights from the CPS. Demographic controls are included for all regressions, but coefficient estimates are omitted here. State and month fixed effects (FEs) are included in all regressions.

Table A5: Robustness to Dropping Observations with Allocation Flags for Outcome Variables – Individual Level

	(1)	(2)	(3)	(4)
	Total Hours Worked		**Hours Actually Worked**	
Parent Type	mothers	fathers	mothers	fathers
% In-Person	-1.095*	0.192	-1.745**	0.070
	(0.659)	(0.585)	(0.733)	(0.690)
% Hybrid	0.471	0.187	0.077	0.100
	(0.538)	(0.444)	(0.595)	(0.530)
Covid Cases Rate	0.046	-0.015	-0.009	0.001
	(0.062)	(0.051)	(0.070)	(0.059)
Month FEs	yes	yes	yes	yes
State FEs	yes	yes	yes	yes
Observations	40,659	42,417	38,921	41,241
R-squared	0.019	0.021	0.017	0.019

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Notes: Table A5 presents the same results as individual-level analysis in Table 7 and Table 8 for mothers and fathers, while dropping observations with allocation flags for the set of 38 states with schooling mode data available. Columns (1) and (2) present the same results as Table 7, while columns (3) and (4) present the same results as Table 8. All regressions are weighted by final person weights from the CPS. Demographic controls are included for all regressions, but coefficient estimates are omitted here. State and month fixed effects (FEs) are included in all regressions.

Table A6: Robustness to Including Industry FEs — Individual Level

	(1)	(2)	(3)	(4)
	Total Hours Worked		**Hours Actually Worked**	
Parent Type	mothers	fathers	mothers	fathers
% In-Person	-1.192*	0.179	-1.767**	0.106
	(0.643)	(0.576)	(0.715)	(0.682)
% Hybrid	0.196	0.265	-0.107	0.195
	(0.527)	(0.440)	(0.584)	(0.525)
Covid Case Rate	0.041	0.003	-0.014	0.011
	(0.061)	(0.051)	(0.068)	(0.058)
Month FEs	yes	yes	yes	yes
State FEs	yes	yes	yes	yes
Industry FEs	yes	yes	yes	yes
Observations	41,146	42,887	39,452	41,783
R-squared	0.045	0.033	0.039	0.030

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Notes: Table A6 presents the same results as individual-level analysis in Table 7 and Table 8 for mothers and fathers when including industry FEs for the set of 38 states with schooling mode data available. This analysis includes controls for the major industry codes from the CPS. Columns (1) and (2) present the same results as Table 7, while columns (3) and (4) present the same results as Table 8. All regressions are weighted by final person weights from the CPS. Demographic controls are included for all regressions, but coefficient estimates are omitted here. State and month fixed effects (FEs) are included in all regressions.

Table A7: Robustness to Including Additional Months — Household Level

	(1)	(2)	(3)	(4)	(5)	(6)
	Total Hours Worked		**Hours Actually Worked**		**Hours Off**	
Household Type	overall	parent	overall	parent	overall	parent
% In-Person	16.704***	19.660***	15.555***	19.054***	2.844	1.526
	(0.691)	(0.967)	(0.740)	(1.038)	(1.773)	(2.295)
% Hybrid	15.545***	17.724***	14.955***	17.086***	0.199	-0.201
	(0.608)	(0.856)	(0.651)	(0.914)	(1.441)	(1.929)
Covid Case Rate	1.135***	1.332***	1.119***	1.215***	0.337	0.400
	(0.094)	(0.133)	(0.101)	(0.142)	(0.292)	(0.372)
State FEs	yes	yes	yes	yes	yes	yes
Month FEs	yes	yes	yes	yes	yes	yes
Observations	84,152	39,676	81,332	38,354	8,453	3,989
R-squared	0.167	0.243	0.140	0.203	0.041	0.057

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Notes: Table A7 replicates the results of Table 10 at the household level extending the analysis to include previously excluded months. All regressions include summer months (August 2020, September 2020, and June 20201) as well as December 2020. All regressions are weighted by the sum of spouses' final person weights from the CPS. Demographic controls are included for all regressions, but coefficient estimates are omitted here. State and month fixed effects (FEs) are included in all regressions.

Table A8: Robustness to Dropping Observations with Allocation Flags for Outcome Variables — Household Level

	(1)	(2)	(3)	(4)
	Total Hours Worked		**Hours Actually Worked**	
Household Type	overall	parent	overall	parent
% In-Person	24.066***	26.849***	22.019***	24.949***
	(0.855)	(1.182)	(0.905)	(1.262)
% Hybrid	20.473***	22.110***	19.234***	21.219***
	(0.729)	(1.017)	(0.781)	(1.086)
Covid Case Rate	1.273***	1.417***	1.167***	1.338***
	(0.120)	(0.169)	(0.129)	(0.187)
State FEs	yes	yes	yes	yes
yes	yes			
Month FEs	yes	yes	yes	yes
yes	yes			
Observations	66,708	31,521	64,483	30,469
R-squared	0.176	0.253	0.146	0.209

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Notes: Table A8 replicates the results of Table 10 at the household level, while dropping observations with allocation flags for the set of 38 states with schooling mode data available. All regressions are weighted by the sum of spouses' final person weights from the CPS. Demographic controls are included for all regressions, but coefficient estimates are omitted here. State and month fixed effects (FEs) are included in all regressions.